PROJECT 1

Adaptive Contrast Restoration for Low-Dose Chest Radiography

Complete Project Report

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Course / Module	Medical signals and image analysis
Software	MATLAB / MATLAB Online
Dataset	NIH ChestX-ray14 - 300 frontal 8-bit PNG radiographs
Project phases	Phase 1: Acquisition Modelling | Phase 2: Enhancement Chain | Phase 3: Evaluation

Project objective

To model low-dose acquisition degradation, develop a spatial-domain enhancement chain, and evaluate whether the enhanced radiographs recover structural and local contrast while controlling noise amplification and false structure.

Project Executive Summary

This project investigates adaptive contrast restoration for simulated low-dose chest radiography using a 300-image working subset of NIH ChestX-ray14. The study is organized into three connected phases: acquisition modelling, enhancement-chain development, and quantitative/visual evaluation.

Phase 1 established the degradation baseline. Bicubic interpolation gave the strongest spatial-resolution reconstruction, 7- and 6-bit quantization remained close to the original images, and false contouring became clearly evident around 4 bits. The Poisson low-dose model produced the expected dose-dependent degradation, with mean SSIM decreasing from 0.9016 at 50% dose to 0.6651 at 10% dose.

Phase 2 evaluated contrast enhancement, denoising, and sharpening methods on Poisson-degraded images. The selected chain was Gaussian denoising (sigma = 1.0), histogram matching to a target distribution derived from the 300 full-dose images, and conservative high-boost filtering (k = 1.00). This chain improved structural similarity at every tested dose level, especially under severe degradation, although pixel-wise error metrics worsened because the enhancement intentionally remapped intensities.

Phase 3 evaluated the selected chain using full-reference metrics, entropy, EME, standardized regional measurements, CLAHE clip-limit sensitivity, and a planned blinded visual-preference study. At 10% dose, mean SSIM increased from 0.6651 to 0.8438 and EME increased from 11.511 to 13.945. CLAHE sensitivity showed noise amplification overtaking contrast gain at the lowest tested clip limit of 0.0025. Numerical evaluation is complete; the remaining project item is the 20-pair visual assessment with at least three human reviewers.

Project Structure

- **Part I - Phase 1: Acquisition Modelling:** Spatial-resolution degradation, intensity quantization, and low-dose Poisson simulation.
- **Part II - Phase 2: Enhancement Chain:** Histogram-based contrast methods, Poisson-noise smoothing, sharpening, final-chain selection, and dose validation.
- **Part III - Phase 3: Evaluation:** Reference/no-reference metrics, regional contrast, CLAHE sensitivity, and structured visual assessment.

PART I

Phase 1 - Acquisition Modelling

Spatial resolution, intensity quantization, and simulated low-dose acquisition

Executive Summary

This report evaluates how three acquisition-related degradations affect chest-radiograph quality: loss of spatial resolution, reduced intensity precision, and reduced photon dose. The analysis was performed on 300 8-bit PNG radiographs from the NIH ChestX-ray14 dataset. Image quality was evaluated using objective similarity/error measures, mean ± standard deviation across the full subset, and representative visual outputs from Images (1), (150), and (300).

Main finding Bicubic interpolation provided the strongest overall reconstruction quality; quantization remained very faithful at 7-6 bits but deteriorated substantially by 4 bits; and simulated dose reduction caused the expected monotonic increase in error and reduction in structural similarity and SNR.

1. Clinical Motivation and Objective

Reducing radiation dose in projection radiography lowers the number of photons reaching the detector. Because photon arrival follows counting statistics, lower exposure increases quantum noise and makes subtle low-contrast structures more difficult to detect. The aim of this project is to characterize these degradation mechanisms before developing a spatial-domain contrast-restoration chain. The restoration stage must improve diagnostic visibility without introducing structures that are not supported by the original image.

1.1 Phase 1 objectives

- Evaluate spatial-resolution loss by downsampling and reconstructing images with nearest-neighbour, bilinear, and bicubic interpolation.
- Evaluate intensity quantization from the original 8-bit images to 7, 6, 5, 4, 3, and 2 bits, with attention to false contouring.
- Simulate reduced-dose radiography using Poisson photon-count noise at 50%, 25%, and 10% relative dose.
- Quantify degradation using MSE, RMSE where applicable, PSNR, SSIM, SNR, and edge-energy retention.

2. Dataset and Experimental Setup

2.1 Dataset

The experiments used a working subset of 300 frontal chest radiographs from NIH ChestX-ray14. The source dataset contains 8-bit PNG chest radiographs at 1024 x 1024 pixels. The working subset was intentionally limited to 300 images so that the experiments could be completed efficiently without downloading the full dataset.

2.2 MATLAB setup

Item	Configuration
Platform	MATLAB
Input folder	Images
Image format	PNG
Number of images	300
Representative images	Image (1), Image (150), Image (300)
Randomness	Fixed random seed for reproducible Poisson simulation

3. Methodology

3.1 Spatial-resolution degradation

Each image was downsampled by factors of 2, 4, and 8. For each factor, three interpolation methods were tested: nearest-neighbour, bilinear, and bicubic. The degraded image was then resized back to the original dimensions and compared with the original image.

Metrics: MSE, PSNR, SSIM, and edge-energy retention. Results were summarized across all 300 images using mean and standard deviation.

3.2 Intensity quantization

The original 8-bit image was retained as the reference. Each image was uniformly requantized to 7, 6, 5, 4, 3, and 2 bits. Lower bit depth reduces the available number of gray levels and can create false contouring or banding, especially in slowly varying regions.

Gray levels by bit depth: 7 bit = 128, 6 bit = 64, 5 bit = 32, 4 bit = 16, 3 bit = 8, and 2 bit = 4 levels.

Metrics: MSE, PSNR, SSIM, and flat-pixel percentage. The 8-bit original is used as the comparison reference and is not treated as another quantized condition.

3.3 Low-dose Poisson simulation

A simplified photon-counting model was used to reproduce the dominant quantum-noise effect expected when dose is reduced. Normalized image intensity was mapped to an approximate expected photon count, Poisson noise was applied, and the result was converted back to normalized image intensity. The simulated relative dose levels were 50%, 25%, and 10%, with the original image retained as the reference.

> **Model assumption** The NIH PNG images are processed display images rather than raw detector measurements. Therefore, the simulated photon counts are approximate and should be interpreted as a controlled model of quantum-noise degradation, not as a scanner-specific dosimetry model.

Low-dose metrics: MSE, RMSE, PSNR, SSIM, and estimated SNR.

4. Results

4.1 Spatial resolution

Table 1 summarizes the 300-image mean and standard deviation for each downsampling factor and interpolation method.

Factor	Method	MSE mean ± SD	PSNR dB mean ± SD	SSIM mean ± SD	Edge retention % mean ± SD
2	Bicubic	4.42E-05 ± 5.98E-05	44.96 ± 2.73	0.9860 ± 0.0038	72.88 ± 5.78
2	Bilinear	7.61E-05 ± 9.94E-05	42.48 ± 2.61	0.9780 ± 0.0059	53.18 ± 6.53
2	Nearest	1.59E-04 ± 1.85E-04	39.18 ± 2.64	0.9609 ± 0.0086	98.33 ± 2.94
4	Bicubic	1.44E-04 ± 2.04E-04	39.87 ± 2.74	0.9622 ± 0.0110	46.58 ± 8.00
4	Bilinear	2.01E-04 ± 2.47E-04	38.14 ± 2.53	0.9529 ± 0.0134	33.15 ± 6.71
4	Nearest	3.43E-04 ± 3.69E-04	35.67 ± 2.45	0.9195 ± 0.0170	123.64 ± 16.06
8	Bicubic	3.43E-04 ± 3.55E-04	35.58 ± 2.33	0.9306 ± 0.0199	26.12 ± 6.46
8	Bilinear	4.51E-04 ± 4.00E-04	34.20 ± 2.15	0.9233 ± 0.0216	18.91 ± 5.05
8	Nearest	7.81E-04 ± 6.05E-04	31.73 ± 2.10	0.8643 ± 0.0266	144.40 ± 28.57

Table 1. Spatial-resolution summary across 300 images.

Bicubic interpolation produced the best overall reconstruction quality at every downsampling factor. At factor 2, bicubic reached a mean SSIM of 0.9860 and PSNR of 44.96 dB. As degradation increased to factor 8, bicubic still retained the highest SSIM (0.9306), while nearest-neighbour fell to 0.8643. This confirms that information loss increases strongly with spatial downsampling and that interpolation choice affects the quality of the reconstructed image.

4.2 Intensity quantization

Bits	Gray levels	MSE mean ± SD	PSNR dB mean ± SD	SSIM mean ± SD	Flat pixels % mean ± SD
7	128	4.78E-06 ± 2.03E-07	53.21 ± 0.18	0.9966 ± 0.0002	14.88 ± 3.43
6	64	2.07E-05 ± 2.51E-07	46.84 ± 0.05	0.9839 ± 0.0008	35.00 ± 5.06
5	32	8.63E-05 ± 1.56E-06	40.64 ± 0.08	0.9458 ± 0.0034	59.49 ± 4.59
4	16	3.68E-04 ± 1.21E-05	34.35 ± 0.14	0.8731 ± 0.0112	78.46 ± 3.07
3	8	1.71E-03 ± 1.20E-04	27.68 ± 0.31	0.8043 ± 0.0275	89.62 ± 1.81
2	4	1.07E-02 ± 1.10E-03	19.72 ± 0.47	0.7598 ± 0.0474	94.85 ± 1.09

Table 2. Quantization summary across 300 images.

Quantization at 7 and 6 bits preserved the original image very well, with mean SSIM values of 0.9966 and 0.9839 respectively. The degradation became more pronounced at 5 bits and increased sharply at 4 bits, where SSIM fell to 0.8731 and the mean flat-pixel percentage increased to 78.46%. At 3 and 2 bits, banding and loss of smooth gray-level transitions are expected to be severe.

False-contouring threshold The quantitative transition is strongest around 4 bits, and the representative images confirm this visually. At 5 bits, anatomy remains recognizable with moderate tonal simplification. At 4 bits, clear banding appears in lung and mediastinal grayscale transitions; at 3 and 2 bits, contouring is severe.

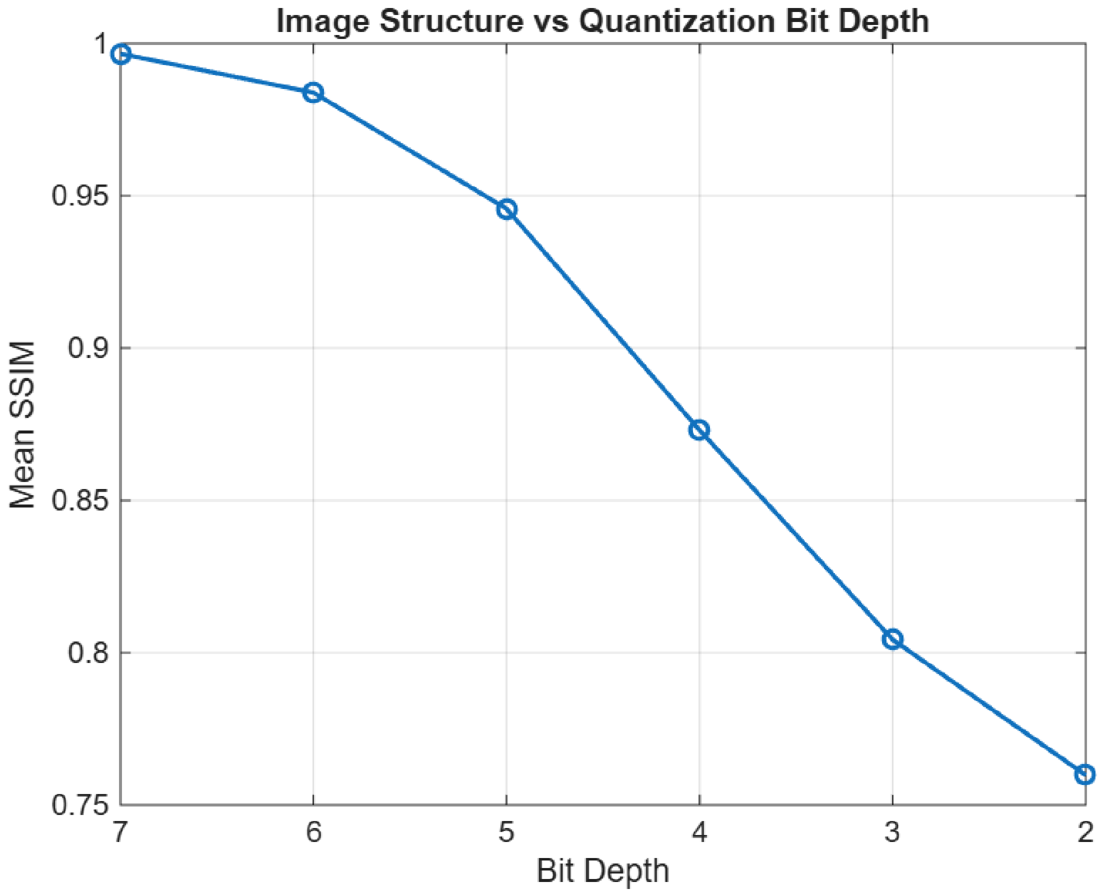

Figure 1. Mean SSIM versus quantization bit depth across the 300-image subset.

4.3 Low-dose simulation

Dose	MSE mean ± SD	RMSE mean ± SD	PSNR dB mean ± SD	SSIM mean ± SD	SNR dB mean ± SD
50%	1.26E-04 ± 1.25E-05	0.01119 ± 0.00059	39.03 ± 0.50	0.9016 ± 0.0114	35.39 ± 0.33
25%	2.51E-04 ± 2.49E-05	0.01583 ± 0.00084	36.03 ± 0.50	0.8244 ± 0.0193	32.38 ± 0.34
10%	6.28E-04 ± 6.17E-05	0.02502 ± 0.00132	32.05 ± 0.49	0.6651 ± 0.0326	28.41 ± 0.34

Table 3. Low-dose simulation summary across 300 images.

The low-dose experiment produced a monotonic decline in image quality as relative dose decreased. Mean SSIM fell from 0.9016 at 50% dose to 0.8244 at 25% and 0.6651 at 10%. At the same time, RMSE increased from 0.01119 to 0.02502, while estimated SNR dropped from 35.39 dB to 28.41 dB.

The error scaling is consistent with a Poisson-noise model. The mean MSE at 25% dose is approximately twice the value at 50% dose, and the mean MSE at 10% dose is approximately five times the 50% value. The SNR falls by approximately 3 dB when the dose is halved from 50% to 25%, which is also consistent with the expected dependence of quantum noise on photon count.

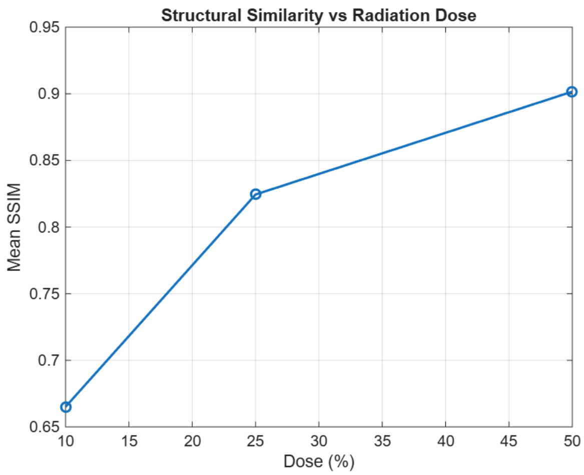

Figure 2. Mean SSIM versus simulated dose.

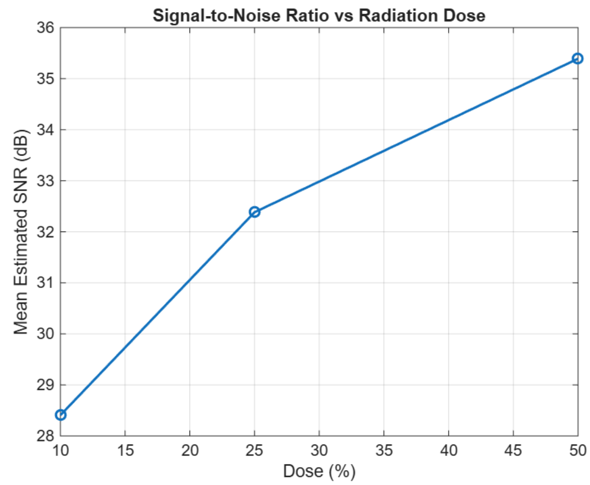

Figure 3. Mean estimated SNR versus simulated dose.

5. Representative Figures

The MATLAB script generated representative outputs for Images (1), (150), and (300). The figures below are the actual saved MATLAB outputs. Together they provide early, middle, and late examples from the 300-image subset, while all quantitative conclusions remain based on the complete 300-image analysis.

5.1 Representative Image (1)

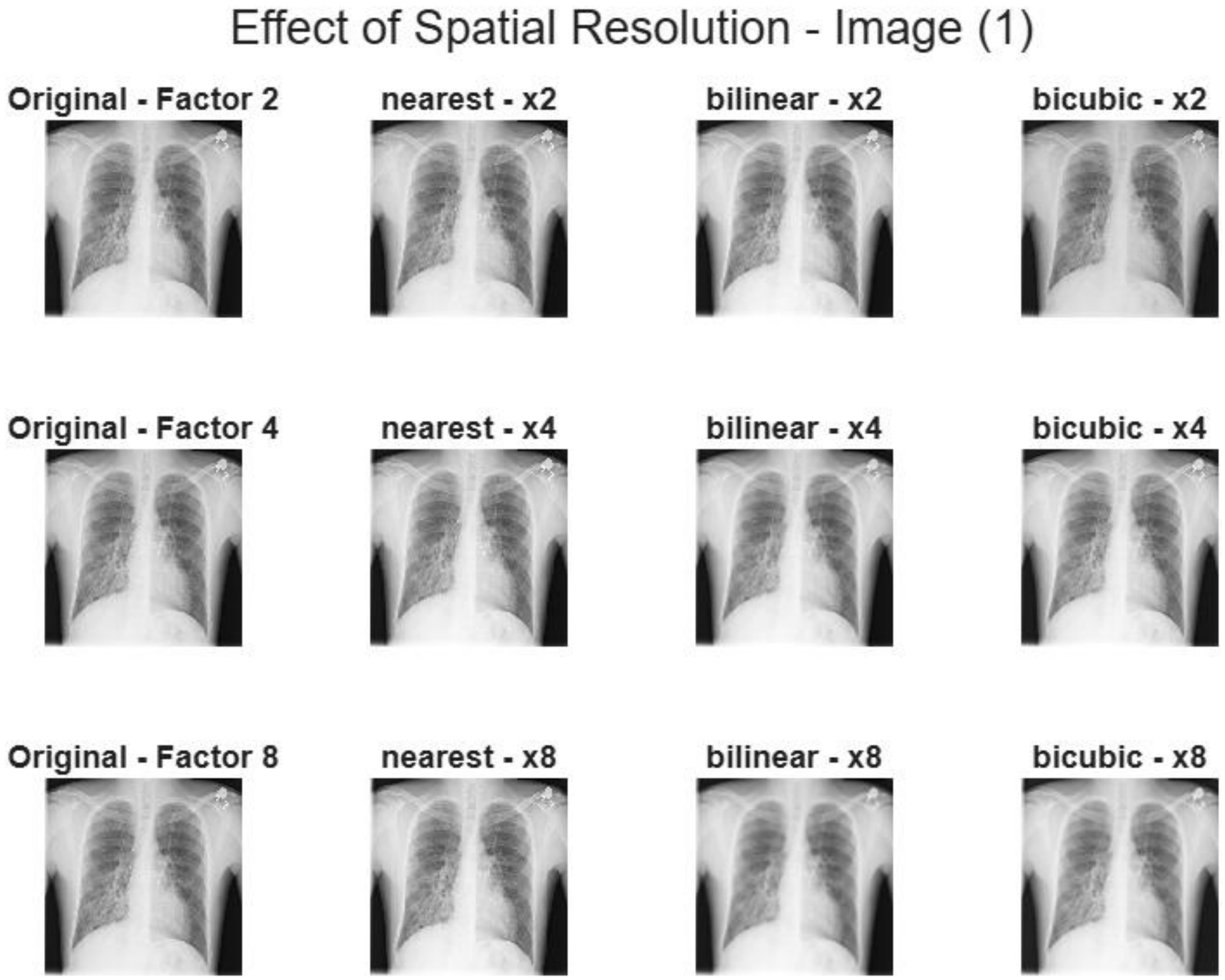

Figure 4. Spatial-resolution comparison for Image (1).

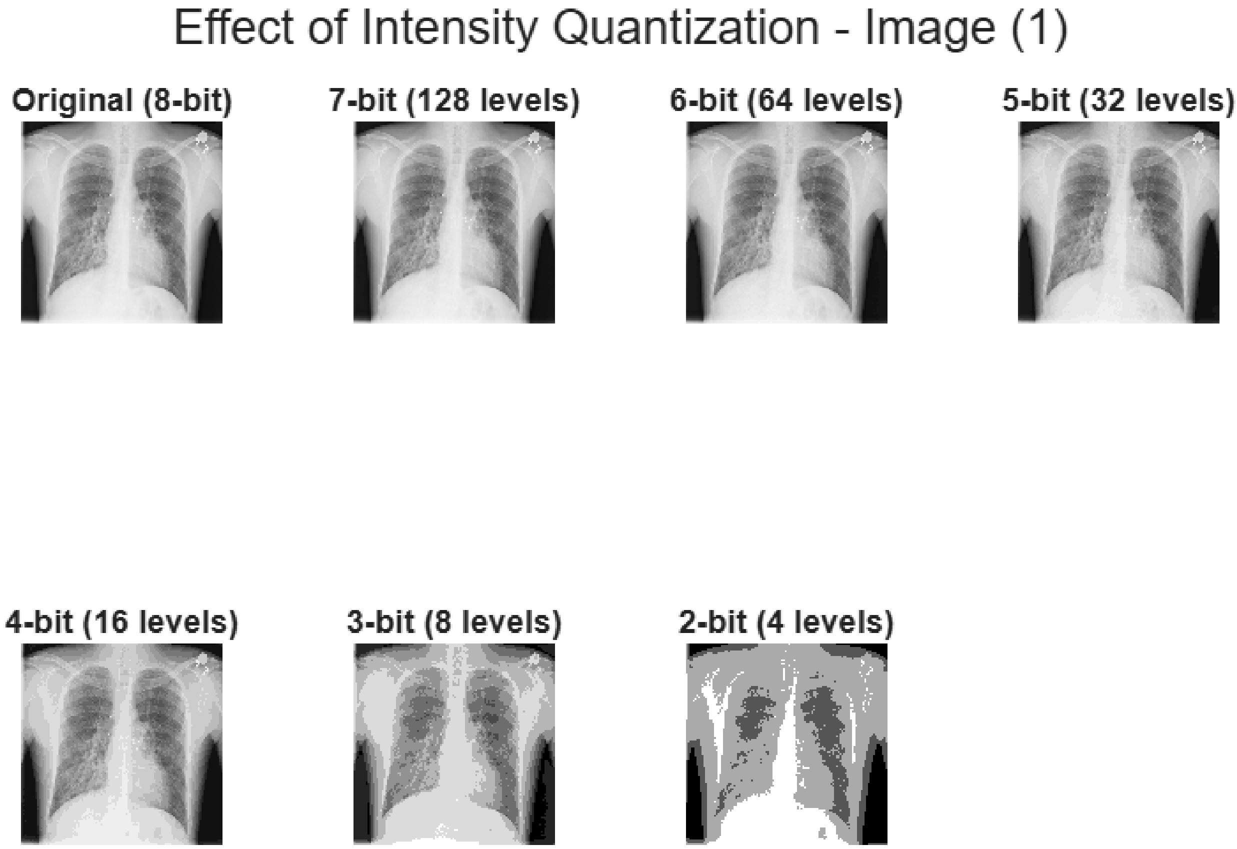

Figure 5. Intensity-quantization comparison for Image (1).

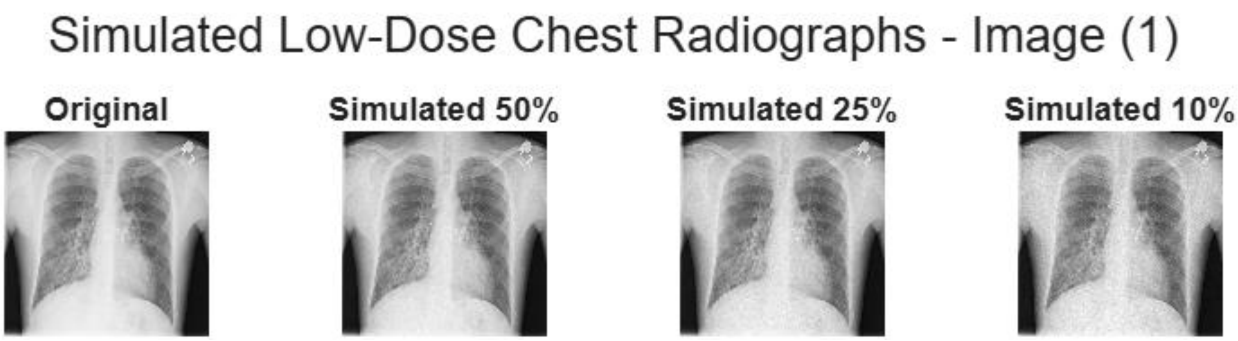

Figure 6. Low-dose Poisson simulation for Image (1).

5.2 Representative Image (150)

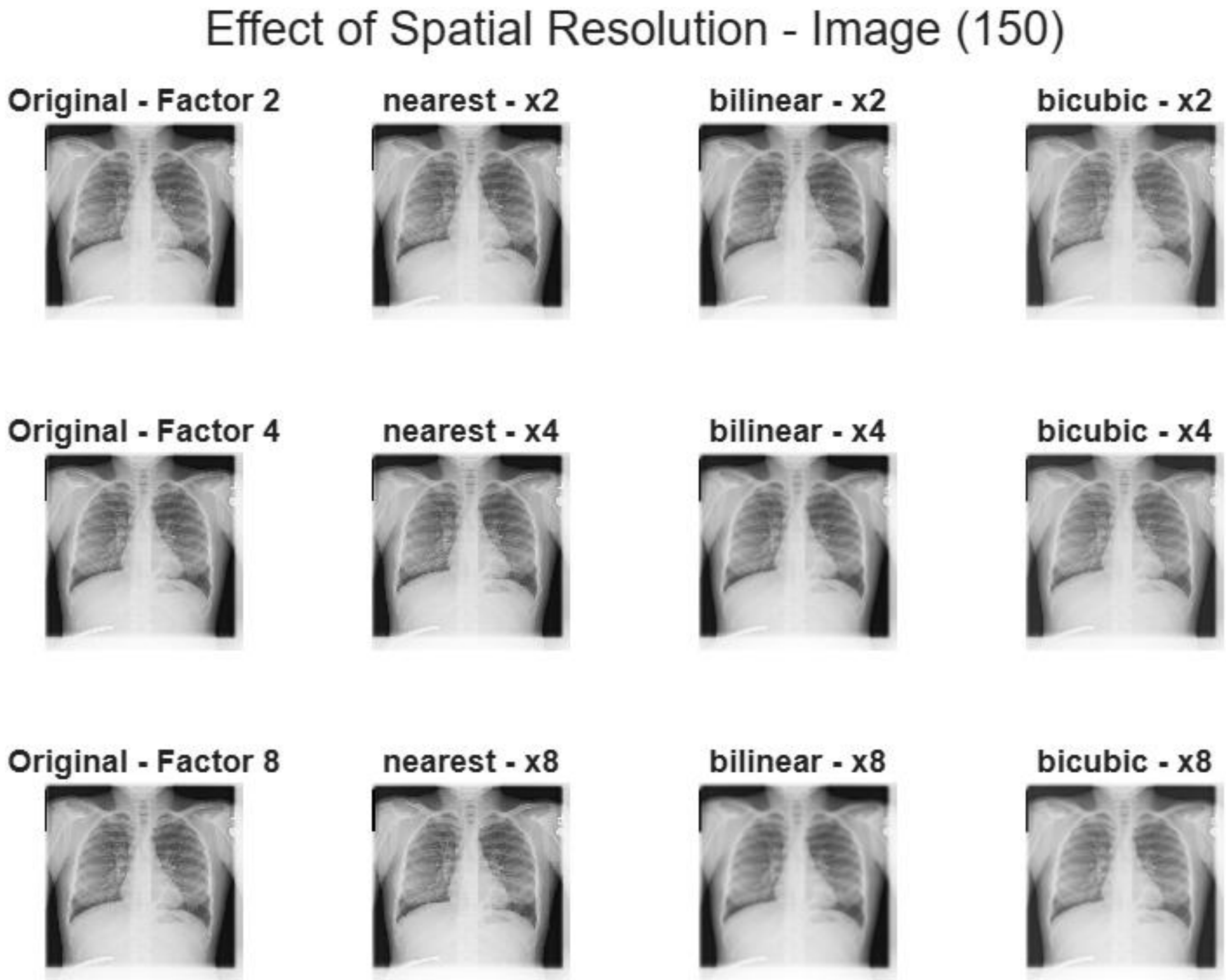

Figure 7. Spatial-resolution comparison for Image (150).

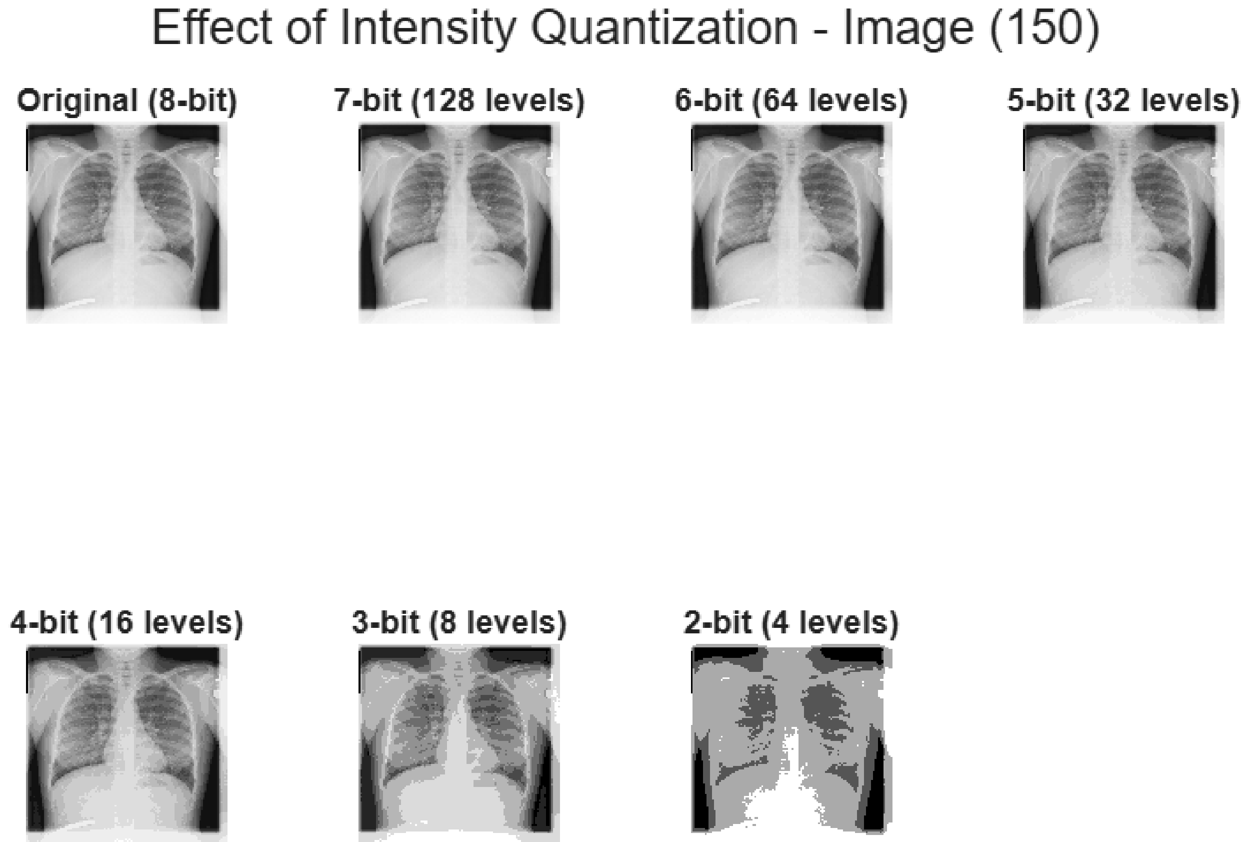

Figure 8. Intensity-quantization comparison for Image (150).

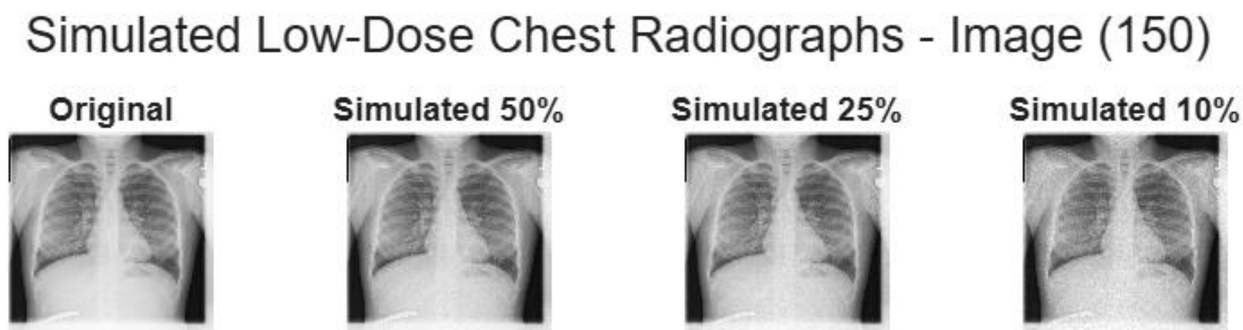

Figure 9. Low-dose Poisson simulation for Image (150).

5.3 Representative Image (300)

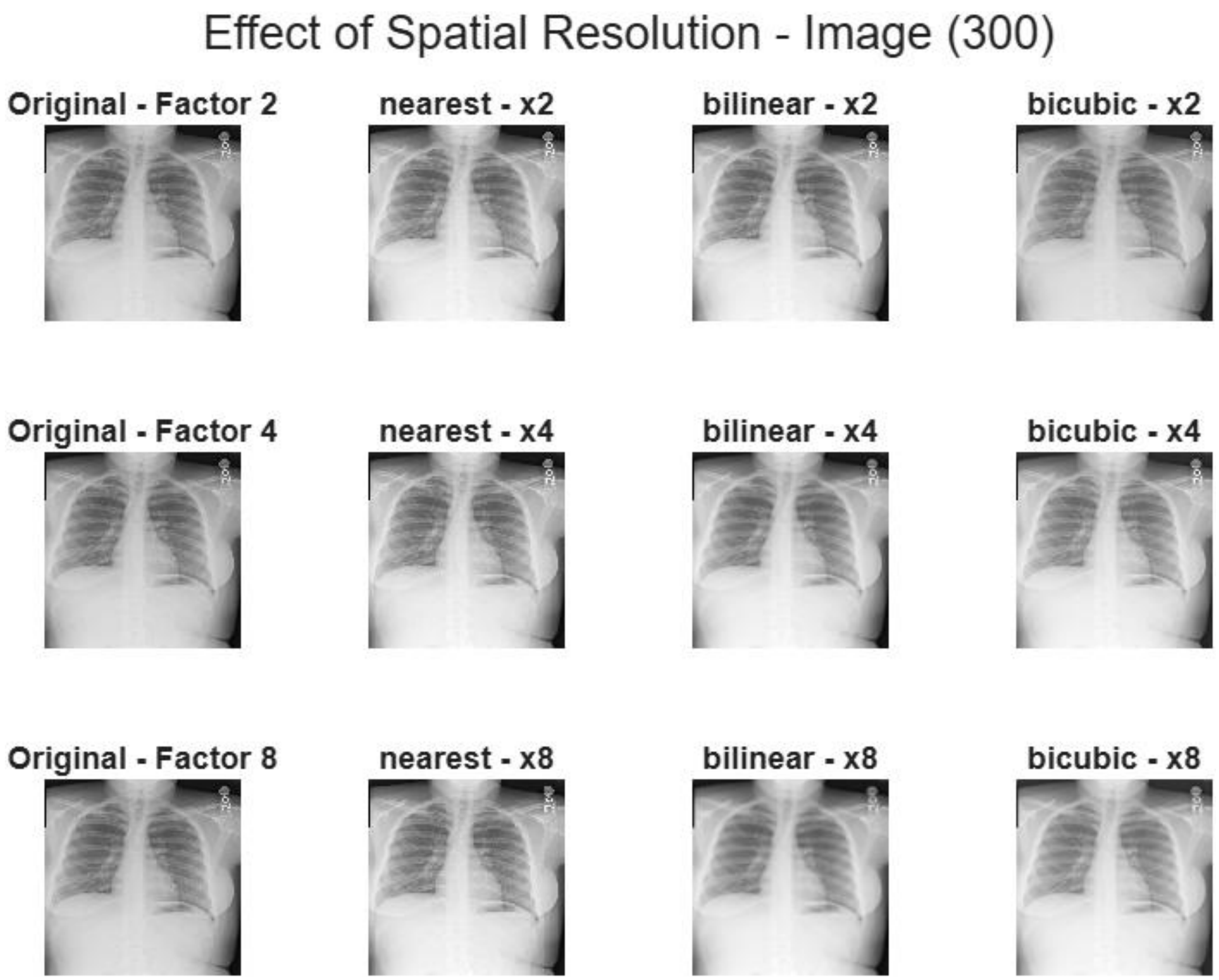

Figure 10. Spatial-resolution comparison for Image (300).

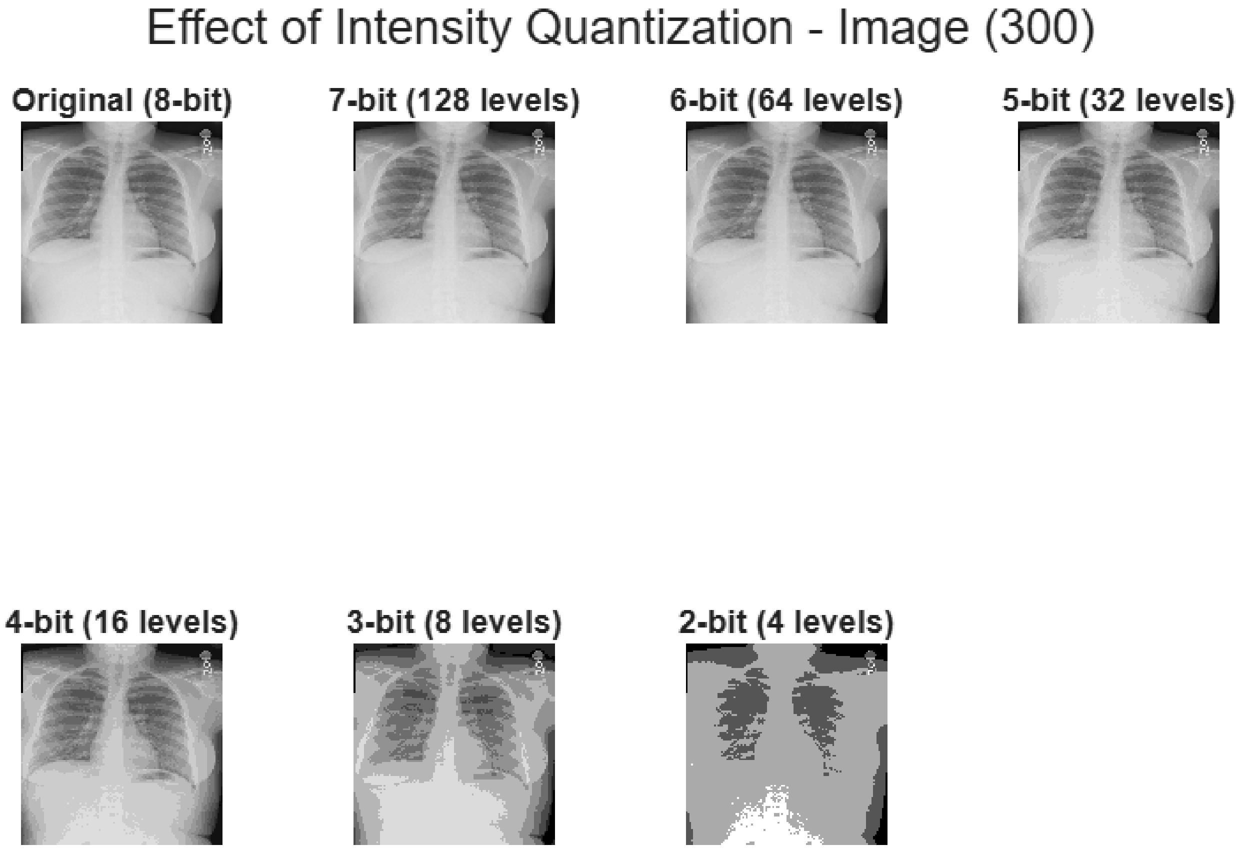

Figure 11. Intensity-quantization comparison for Image (300).

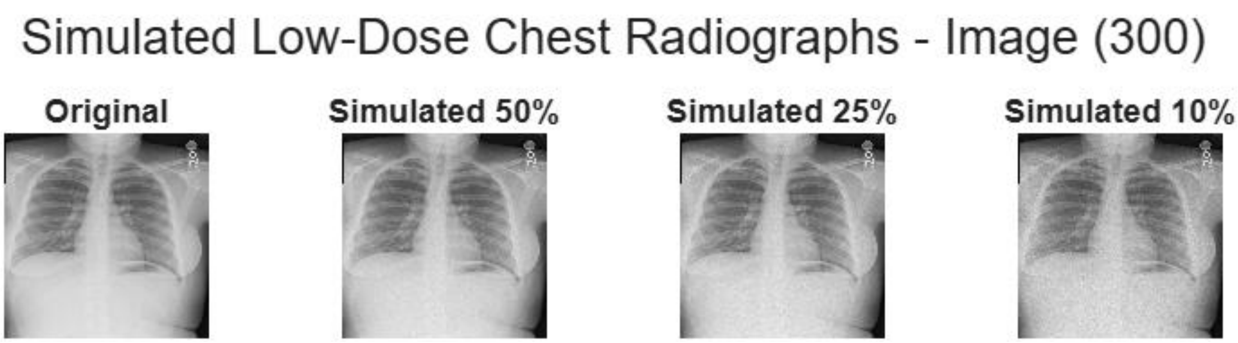

Figure 12. Low-dose Poisson simulation for Image (300).

6. Discussion

6.1 Spatial-resolution findings

The results demonstrate that spatial downsampling causes irreversible loss of fine information. Even though the image is resized back to 1024 x 1024, the original high-frequency detail cannot be recreated. Bicubic interpolation consistently achieved the best balance of low error and high structural similarity. Bilinear was intermediate, while nearest-neighbour produced the lowest fidelity and visible blockiness at stronger degradation.

6.2 Quantization findings

The quantization experiment shows that chest radiographs tolerate a moderate reduction in gray-level precision before strong structural degradation appears. Seven- and six-bit images remain visually close to the original and retain very high SSIM. At five bits, tonal simplification becomes more noticeable while the main anatomical structures remain well represented. At four bits, both the large increase in flat-pixel percentage and the representative images show obvious false contouring in broad lung-field and mediastinal intensity transitions. The 3- and 2-bit examples show severe posterization, confirming that these bit depths do not preserve diagnostically smooth grayscale information.

6.3 Low-dose findings

The low-dose simulation behaves as expected for quantum-noise-limited imaging. Lower dose reduces the expected photon count and increases relative Poisson fluctuations. The representative images show progressively stronger grain/noise from 50% to 25% and 10% dose, matching the monotonic decline in SSIM and SNR. At 50% dose, structural similarity remains relatively high; at 25%, noise becomes more evident; and at 10%, image fidelity is substantially degraded. This establishes a controlled low-dose input for the enhancement chain developed in Phase 2.

7. Limitations

- The NIH ChestX-ray14 PNGs are processed 8-bit images, not raw detector data. Absolute detector photon counts and acquisition parameters are not available.
- The low-dose simulation therefore models the statistical effect of reduced photon counts rather than a specific X-ray system or exact clinical exposure setting.
- Global metrics such as MSE, PSNR, and SSIM do not directly measure diagnostic performance. A high metric value does not guarantee that every subtle lesion remains visible.
- The approximately 4-bit false-contouring threshold is supported by both the numerical trend and the representative visual outputs. Across Images (1), (150), and (300), the 4-bit images show clear loss of smooth gray-level transitions, while 3- and 2-bit images show severe banding and posterization.
- Nearest-neighbour edge-energy values greater than 100% reflect artificial pixel-block boundaries and should not be interpreted as true anatomical information retention.

8. Conclusion

Phase 1 established a quantitative baseline for the acquisition degradations that the later restoration stages must address. Bicubic interpolation was the strongest reconstruction method across all spatial downsampling factors. Intensity quantization was well tolerated at 7 and 6 bits but degraded rapidly below 5 bits, with 4 bits marking the approximate transition to clearly visible contouring. The Poisson low-dose model produced the expected dose-dependent increase in error and loss of structural similarity: 50% dose retained a mean SSIM of 0.902, whereas 10% dose fell to 0.665. These results provide a rigorous basis for designing and evaluating the adaptive contrast-restoration stage of the project.

9. Reproducibility Notes

- Input images: PNG files stored in the MATLAB folder named Images.
- Dataset size used in the experiment: 300 images.
- Spatial factors: 2, 4, 8.
- Interpolation methods: nearest-neighbour, bilinear, bicubic.
- Quantization levels: 7, 6, 5, 4, 3, 2 bits compared with the original 8-bit image.
- Low-dose simulations: 50%, 25%, 10% relative dose, compared with the original image.
- Representative visual outputs: Image (1), Image (150), and Image (300).
- All summary values are reported as mean and standard deviation across the 300-image subset.

PART II

Phase 2 - Enhancement Chain

Denoising, contrast restoration, sharpening, and final-chain validation

1. Phase 2 Objective

Phase 2 evaluates spatial-domain enhancement methods for radiographs degraded by the Poisson low-dose model established in Phase 1. The objective is not simply to maximize visible contrast. The enhancement chain must improve structural fidelity and low-contrast visibility while avoiding noise amplification, halos, block artifacts, and false anatomical structures.

Phase 2 selected chain: Gaussian denoising (σ = 1.0) → Histogram matching → High-boost filtering (k = 1.00). Validated at 50%, 25%, and 10% simulated dose across the 300-image working subset.

2. Dataset and Baseline

The Phase 2 analysis used 300 frontal NIH ChestX-ray14 radiographs stored as 8-bit PNG images. The original full-dose PNG image was retained as the reference for quantitative evaluation. The same Poisson low-dose model established in Phase 1 was used, so the smoothing and enhancement stages were evaluated on Poisson-degraded radiographs rather than on synthetically added Gaussian noise.

Phase 2 low-dose baseline for the 300-image run

Dose	Mean MSE ± SD	Mean RMSE ± SD	Mean PSNR ± SD	Mean SSIM ± SD	Mean SNR ± SD
50%	0.00012564 ± 0.00001247	0.01119 ± 0.00059	39.03 ± 0.50 dB	0.9016 ± 0.0114	35.39 ± 0.33 dB
25%	0.00025123 ± 0.00002486	0.01583 ± 0.00084	36.03 ± 0.50 dB	0.8244 ± 0.0193	32.38 ± 0.34 dB
10%	0.00062753 ± 0.00006168	0.02502 ± 0.00132	32.05 ± 0.49 dB	0.6651 ± 0.0326	28.41 ± 0.34 dB

Table 1. Baseline low-dose results measured across the 300-image Phase 2 subset before enhancement.

3. Experimental Design

A 25% simulated dose condition was used as the primary development condition because it provides substantial but non-extreme Poisson degradation. Candidate operations were evaluated at the 25% development dose, and the selected final chain was then evaluated across the 300-image subset at 50%, 25%, and 10% dose.

- Quantitative summaries are reported as mean ± standard deviation across the 300 images.
- Representative visual comparisons are shown for Image (1), Image (150), and Image (300).
- The original 8-bit full-dose image remains the reference for MSE, RMSE, PSNR, and SSIM.
- Method selection prioritizes structural fidelity and controlled enhancement rather than visual contrast alone.

4. Contrast Enhancement

4.1 Global Histogram Equalization

Global histogram equalization remaps the image gray levels using the cumulative distribution function of the entire image. It is evaluated because it can expand global contrast, but in low-dose radiographs it can also amplify quantum noise and overemphasize already high-contrast structures. The method is therefore judged using structural metrics as well as visual inspection.

4.2 Histogram Matching

Histogram matching maps each degraded image toward a target gray-level distribution derived from the original full-dose images. In this run, the target histogram was computed from all 300 full-dose reference radiographs. The resulting distribution is broad and multimodal, with a dominant high-intensity region around approximately 190–215 gray levels and smaller low-intensity modes. Using a dataset-derived target reduces dependence on a single patient image and was selected as the contrast stage of the final chain.

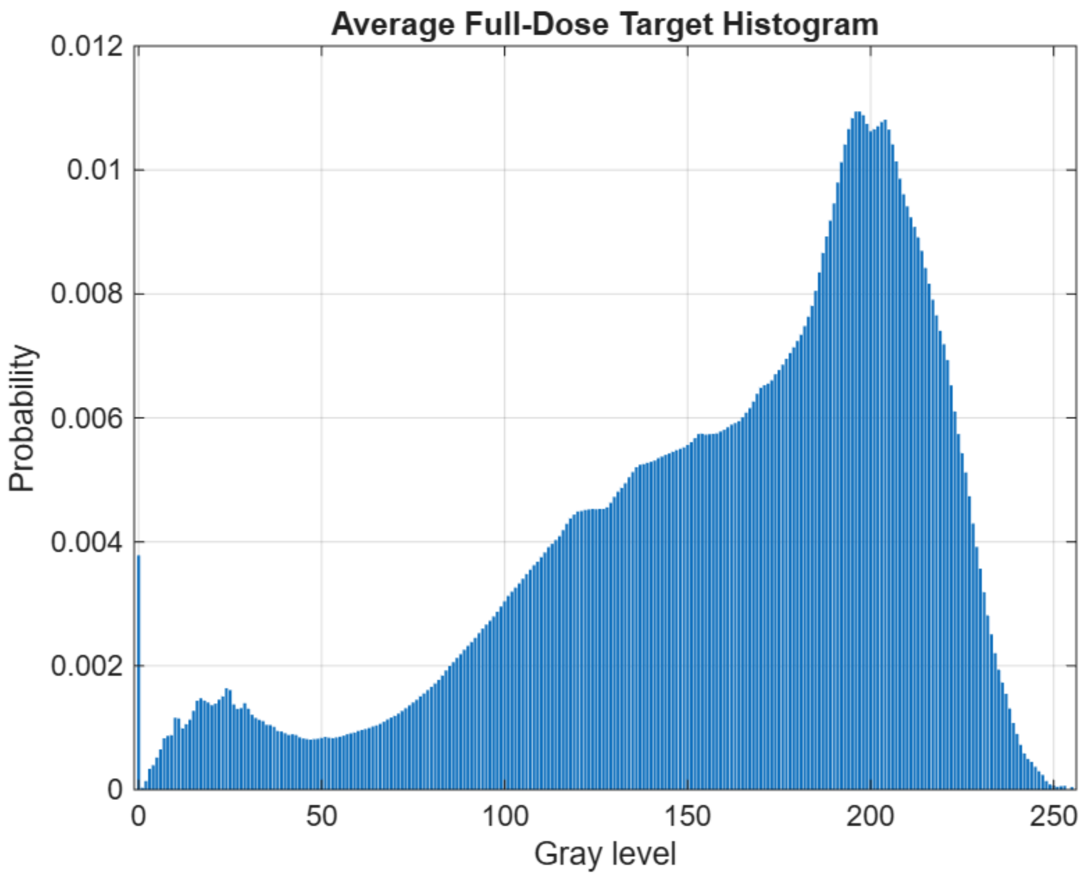

Figure 1. Average full-dose target histogram derived from the 300 original radiographs.

4.3 Adaptive Histogram Equalization (AHE)

AHE performs histogram equalization locally on a tile grid. Candidate tile grids are compared because a small tile size can reveal local contrast but may strongly amplify Poisson noise, whereas a large tile grid behaves more like global equalization. The experiment therefore evaluates the local-contrast/noise trade-off rather than assuming that smaller tiles are always better.

4.4 Custom Contrast-Limited AHE (CLAHE)

CLAHE is implemented with a custom clipping and redistribution stage as required by the project. For each local tile, the histogram is calculated, bins above the selected clip threshold are clipped, and the removed pixel count is accumulated as excess. The excess is redistributed across the histogram bins before the cumulative distribution function is calculated and used for gray-level mapping. Interpolation between neighboring tile mappings is used to suppress visible tile boundaries.

Custom CLAHE step	Implementation purpose
1. Tile histogram	Estimate the local intensity distribution.
2. Clip histogram	Limit dominant bins so local contrast enhancement cannot grow without control.
3. Accumulate excess	Count pixels removed by histogram clipping.
4. Redistribute excess	Spread clipped counts back across gray-level bins.
5. Compute local CDF	Create the local intensity mapping.
6. Interpolate mappings	Reduce discontinuities between adjacent tiles.

Table 2. Custom CLAHE clipping and redistribution workflow.

5. Smoothing on Poisson Noise

All smoothing methods are applied to Poisson-degraded images. No Gaussian noise is substituted for the low-dose degradation. This is important because filter behavior is noise-model dependent.

Box filter: Local arithmetic averaging. Strong noise reduction is possible, but fine edges and thin structures may be blurred.

Gaussian filter: Weighted spatial averaging. It generally produces smoother transitions than a box filter but can still blur fine anatomical detail.

Median filter: Nonlinear neighborhood median. It is particularly effective for impulse noise, so its usefulness for Poisson noise must be established experimentally rather than assumed.

Adaptive median filter: The neighborhood size changes according to local statistics. The method attempts to distinguish corrupted samples from legitimate structure while limiting unnecessary smoothing.

Alpha-trimmed mean filter: Neighborhood values are sorted, a selected number of the highest and lowest values are removed, and the remaining values are averaged. This combines aspects of mean and median filtering.

6. Sharpening

6.1 Laplacian Sharpening

Laplacian sharpening emphasizes second-order intensity transitions. It can enhance boundaries but is sensitive to residual noise, so it is evaluated after denoising rather than directly on the raw Poisson-degraded image.

6.2 Unsharp Masking

Unsharp masking forms a detail mask by subtracting a blurred image from the input and adds a controlled fraction of that mask back to the image. This allows high-frequency detail to be restored after smoothing while reducing the risk of strong second-derivative noise amplification.

6.3 High-Boost Filtering

High-boost filtering extends unsharp masking by sweeping the boost factor. The parameter sweep is necessary because a low boost may provide insufficient detail recovery, whereas a high boost can generate halos and amplify residual Poisson noise. The final boost factor is selected from the dataset-level quantitative results and representative images.

7. Quantitative Evaluation

The main reference-based metrics are MSE, RMSE, PSNR, and SSIM, calculated against the original full-dose image. Entropy and edge-related measures may be used as supporting descriptors, but they are not treated as stand-alone indicators of diagnostic quality. A method that increases entropy or edge magnitude by amplifying noise is not automatically considered superior.

Metric	Interpretation in Phase 2
MSE	Average squared reconstruction error; lower is better.
RMSE	Error expressed on the image intensity scale; lower is better.
PSNR	Signal fidelity relative to reconstruction error; higher is better.
SSIM	Structural similarity to the original full-dose radiograph; higher is better.
Entropy	Supporting descriptor of intensity information; can also rise due to noise.
Edge retention	Supporting descriptor of high-frequency content; values must be interpreted for artificial edge amplification.

Table 3. Evaluation metrics and interpretation.

8. Results

The Phase 2 workflow was applied to the 300-image working subset, with the final selected chain evaluated at all three dose levels. The supplied final-chain summary and representative figures show that the selected chain was Gaussian denoising ($\sigma = 1.0$) → histogram matching → high-boost filtering ($k = 1.00$). Detailed per-method sweep tables for every candidate contrast, smoothing, and sharpening setting were not included in the supplied output; therefore, those comparisons are reported only to the extent supported by the available results.

8.1 Selected Contrast Stage

Stage / Method	Selected parameters	Evidence available	Quantitative stage metrics	Interpretation
Contrast	Histogram matching	Selected in final chain; target histogram supplied	Per-stage sweep summary not supplied	Restores a full-dose-like global tonal distribution.
Target distribution	Average full-dose histogram	Derived from 300 original radiographs	Histogram figure supplied	Broad multimodal target; dominant high-intensity region around 190–215 gray levels.

Stage / Method	Selected parameters	Evidence available	Quantitative stage metrics	Interpretation
Other candidates	Global HE / AHE / custom CLAHE	Methods described in implementation	Detailed comparative summary not supplied	No numerical ranking is claimed in this report.
Final role	Contrast restoration	Placed after denoising	Final-chain metrics reported separately	Balances tonal recovery with structural preservation.

Table 4. Selected contrast stage and available evidence.

8.2 Selected Smoothing Stage on Poisson Noise

Stage / Method	Selected parameters	Evidence available	Quantitative stage metrics	Behavior on Poisson noise
Denoising	Gaussian	σ = 1.0	Per-stage sweep summary not supplied	Representative images show reduced grain with preserved gross anatomy.
Other candidates	Box / median / adaptive median / alpha-trimmed mean	Implemented in study	Detailed comparative summary not supplied	No unsupported numerical ranking is reported.
Noise model	Poisson	25% dose used for development	Baseline metrics supplied	Filter behavior is evaluated on quantum-noise degradation.
Final role	Pre-contrast denoising	First stage of chain	Final-chain metrics reported separately	Reduces noise before histogram and sharpening operations.
Selection scope	300-image working subset	Dataset-level evaluation	300 images	Selected chain evaluated across all 300 images.

Table 5. Selected denoising stage and available evidence.

8.3 Selected Sharpening Stage

Stage / Method	Selected parameters	Evidence available	Quantitative stage metrics	Artifact / detail observation
Sharpening	High-boost	k = 1.00	Per-stage sweep summary not supplied	Conservative sharpening; no obvious large halo in supplied figures.
Other candidates	Laplacian / unsharp	Implemented in study	Detailed comparative summary not supplied	No numerical ranking is claimed.

Stage / Method	Selected parameters	Evidence available	Quantitative stage metrics	Artifact / detail observation
Placement	After denoising + contrast	Final stage	Final-chain metrics supplied	Avoids directly sharpening raw Poisson noise.

Table 6. Selected sharpening stage and available evidence.

8.4 Final Enhancement Chain and Dose Validation

Dose	Baseline SSIM	Final SSIM ± SD	Baseline PSNR	Final PSNR ± SD	Baseline RMSE	Final RMSE ± SD
50%	0.9016 ± 0.0114	0.9093 ± 0.0442	39.03 ± 0.50 dB	25.68 ± 4.98 dB	0.01119 ± 0.00059	0.06254 ± 0.04719
25%	0.8244 ± 0.0193	0.8904 ± 0.0455	36.03 ± 0.50 dB	25.62 ± 4.93 dB	0.01583 ± 0.00084	0.06283 ± 0.04710
10%	0.6651 ± 0.0326	0.8438 ± 0.0492	32.05 ± 0.49 dB	25.44 ± 4.79 dB	0.02502 ± 0.00132	0.06364 ± 0.04684

Table 7. Final-chain validation at 50%, 25%, and 10% simulated dose.

Final selected chain: Gaussian σ=1.0 → Histogram matching → High-boost k=1.00

9. Representative Figures

The representative figures below show Image (1), Image (150), and Image (300). Each sequence contains the original full-dose reference, the 25% Poisson-degraded image, Gaussian denoising with σ = 1.0, histogram matching, and the final high-boost result with k = 1.00. Visual inspection focuses on preservation of lung markings and boundaries, suppression of quantum-grain texture, tonal recovery, and avoidance of obvious halos or false structure.

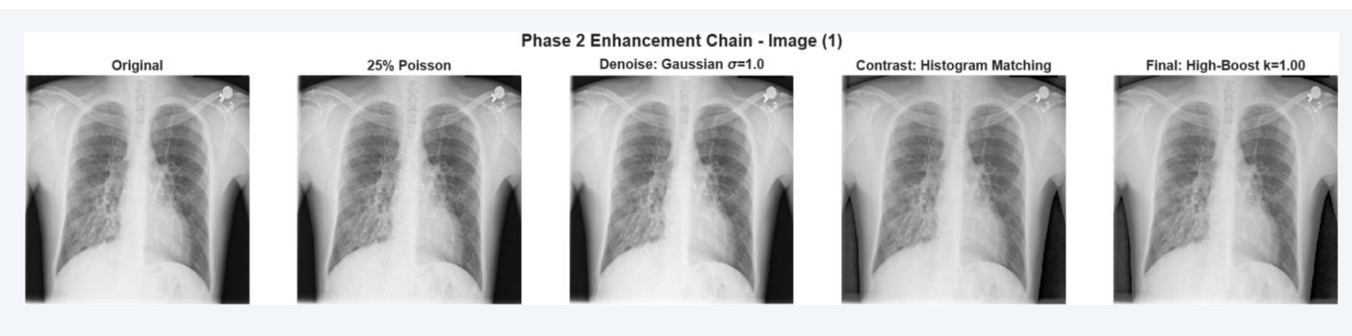

Figure 2. Phase 2 enhancement chain for Image (1).

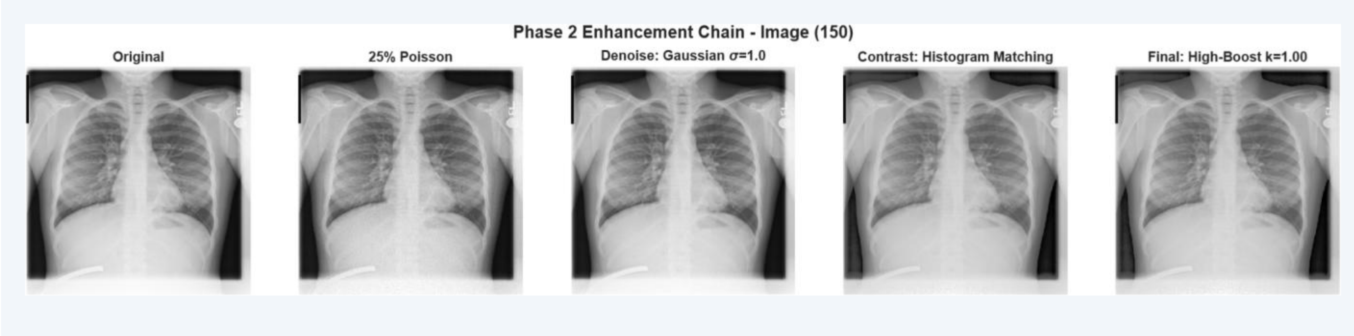

Figure 3. Phase 2 enhancement chain for Image (150).

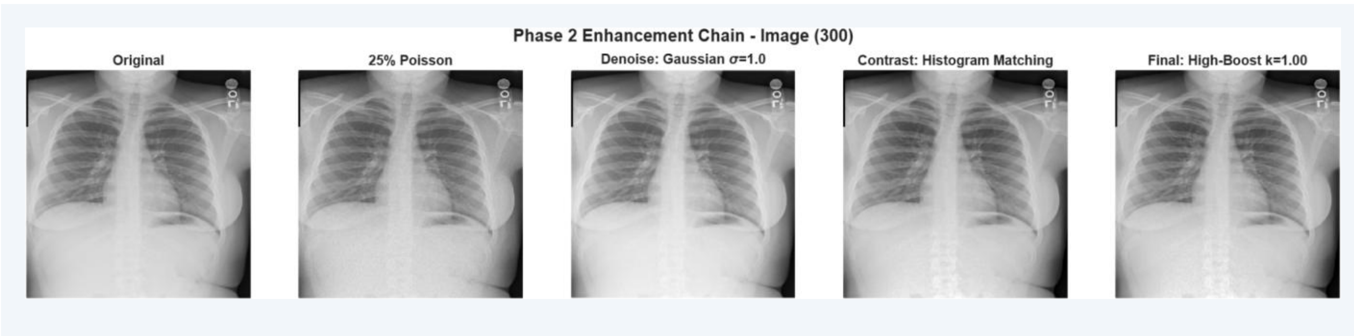

Figure 4. Phase 2 enhancement chain for Image (300).

10. Discussion

The selected chain improved structural similarity at all three tested dose levels, but it did not improve every fidelity metric. This distinction is important when interpreting enhancement of medical images.

- At 50% dose, mean SSIM increased from 0.9016 ± 0.0114 for the baseline to 0.9093 ± 0.0442 after enhancement. At 25% dose, SSIM increased from 0.8244 ± 0.0193 to 0.8904 ± 0.0455. The largest structural improvement occurred at 10% dose, where SSIM increased from 0.6651 ± 0.0326 to 0.8438 ± 0.0492.
- Despite the SSIM improvement, MSE, RMSE, PSNR, and SNR became worse after the final chain. This is consistent with the fact that histogram matching deliberately changes the gray-level distribution and high-boost filtering modifies local intensities. The enhanced image can therefore become structurally more similar while becoming less faithful in a pixel-by-pixel radiometric sense.
- The 25% development case illustrates the trade-off clearly across 300 images: SSIM increased from 0.8244 to 0.8904, whereas PSNR decreased from 36.03 dB to 25.62 dB and RMSE increased from 0.01583 to 0.06283. The enhancement chain should therefore be described as structure- and contrast-oriented rather than as a minimum-error reconstruction.
- The target full-dose histogram provides a dataset-level tonal reference. The representative figures indicate that histogram matching moves the degraded/denoised images toward a full-dose-like global appearance without the aggressive contrast expansion that can occur with unconstrained global histogram equalization.
- Gaussian smoothing with σ = 1.0 was selected as the denoising stage. In the representative 25% dose images, it visibly reduces the fine Poisson-grain texture while retaining the main anatomical boundaries. Because only the selected-stage output was supplied, the report does not claim that Gaussian smoothing outperformed every other tested filter by a specific numerical margin.
- High-boost filtering with k = 1.00 was selected as the final sharpening stage. In the supplied representative images, the sharpening is mild and does not produce obvious large halos at the displayed scale. The chosen

factor is conservative, which is appropriate because aggressive sharpening can amplify residual Poisson noise or create false edge structure.
- Edge-retention percentages are above 100% for several baseline and enhanced conditions. These values should not be interpreted as recovery of additional true anatomy. Poisson noise and sharpening both increase gradient energy, so edge retention is treated only as a supporting descriptor and not as the principal quality criterion.

11. Justification of Operation Order

The measured final chain supports the proposed ordering: Gaussian denoising (σ = 1.0) → histogram matching → high-boost filtering (k = 1.00). Denoising first limits propagation of Poisson noise into later stages. Histogram matching then restores the global tonal distribution toward the full-dose reference, and conservative high-boost filtering is applied last to recover edge crispness after smoothing.

Measured ordering conclusion: The selected result supports denoising → contrast restoration → sharpening: Gaussian σ=1.0 → histogram matching → high-boost k=1.00.

12. Limitations

The NIH ChestX-ray14 PNG files are processed 8-bit images rather than raw detector acquisitions, so the Poisson model remains an approximate educational simulation. In addition, the analysis uses a 300-image working subset rather than the complete NIH ChestX-ray14 collection. The results therefore characterize this selected subset and should not be interpreted as a population-level diagnostic-performance estimate. Reference-based metrics also measure similarity to the original image rather than diagnostic accuracy. Finally, the supplied output contains the final-chain summary and selected-stage figures, but not the full per-method sweep tables for all contrast, smoothing, and sharpening candidates.

13. Conclusion

For the 300-image Phase 2 subset, the selected enhancement chain was Gaussian denoising (σ = 1.0) → histogram matching → high-boost filtering (k = 1.00). Across the full subset, the chain improved mean SSIM at all three dose levels, with the largest gain at 10% dose (0.6651 to 0.8438). However, RMSE increased and PSNR decreased because histogram matching and high-boost filtering intentionally modify image intensities. The final chain is therefore best justified as a controlled structure/contrast-restoration strategy rather than a pixel-faithful minimum-error reconstruction.

Final Phase 2 conclusion: Across the 300-image subset, Gaussian σ=1.0 → histogram matching → high-boost k=1.00 improved mean SSIM at all dose levels, with the strongest gain at 10% dose (0.6651 → 0.8438). The accompanying decrease in PSNR and increase in RMSE show that the method performs deliberate tonal and local-detail remapping rather than minimum-error reconstruction.

14. Reproducibility Notes

- Input folder: Images
- Input format: PNG
- Working subset: 300 images
- Representative figures: Images 1, 150, and 300

- Primary development condition: 25% simulated dose
- Final-chain verification: 50%, 25%, and 10% simulated dose
- Summary reporting: mean ± standard deviation across 300 images
- Custom requirement: CLAHE clipping and histogram redistribution implemented directly in MATLAB code

PART III

Phase 3 - Evaluation

Full-reference, no-reference, regional, sensitivity, and visual assessment

1. Executive Summary

Phase 3 evaluates whether the enhancement pipeline developed in Phase 2 improves the perceptual and structural quality of simulated low-dose chest radiographs without simply amplifying noise. The evaluation was performed on 300 chest radiographs at simulated dose levels of 50%, 25%, and 10%. Full-reference metrics were calculated against the corresponding original full-dose images, while entropy and the measure of enhancement (EME) were used as no-reference indicators of information spread and local contrast.

Across all three dose levels, enhancement increased mean SSIM substantially, with the largest gain at 10% dose (0.6651 to 0.8438). Entropy and EME also increased after enhancement, indicating a broader intensity distribution and stronger local contrast. However, RMSE increased and PSNR decreased after enhancement. This shows that the final chain does not minimize pixel-wise error; rather, it deliberately remaps intensities and local contrast to recover structural appearance. The result therefore represents a trade-off between fidelity to the exact original pixel values and perceptual/structural restoration.

Key finding The enhancement chain provides its clearest structural benefit under severe dose degradation. At 10% dose, SSIM improved by about 26.9%, while EME increased by about 21.1%.

Important limitation The assignment requested manually placed ROIs. This implementation used standardized automatic geometric ROIs for left lung, right lung, and mediastinum. The report therefore treats ROI findings as reproducible regional sampling rather than manually annotated anatomy.

2. Evaluation Design

2.1 Full-reference metrics

Each enhanced or unenhanced low-dose image was compared with its original full-dose reference. Three metrics were used: RMSE, PSNR, and SSIM. RMSE quantifies pixel-wise error, PSNR expresses error on a logarithmic decibel scale, and SSIM emphasizes structural similarity in luminance, contrast, and local structure. Lower RMSE and higher PSNR/SSIM indicate closer agreement with the reference, but enhancement algorithms that intentionally alter contrast may improve SSIM while worsening RMSE and PSNR.

2.2 No-reference metrics

Entropy was used to describe the spread of image intensity information, while EME quantified local enhancement by dividing the image into blocks and measuring local max-to-min contrast. Higher values can indicate stronger contrast, but they do not by themselves prove diagnostic improvement because noise can also increase entropy and local contrast.

2.3 Regional contrast assessment

To avoid interactive ROI selection, fixed proportional ROIs were placed automatically over the left lung, right lung, and mediastinal regions. The same geometric definitions were applied consistently to the representative images. This improves repeatability but does not provide true anatomical segmentation, and the boxes may include vessels, ribs, cardiac silhouette, or non-lung tissue depending on patient positioning and anatomy.

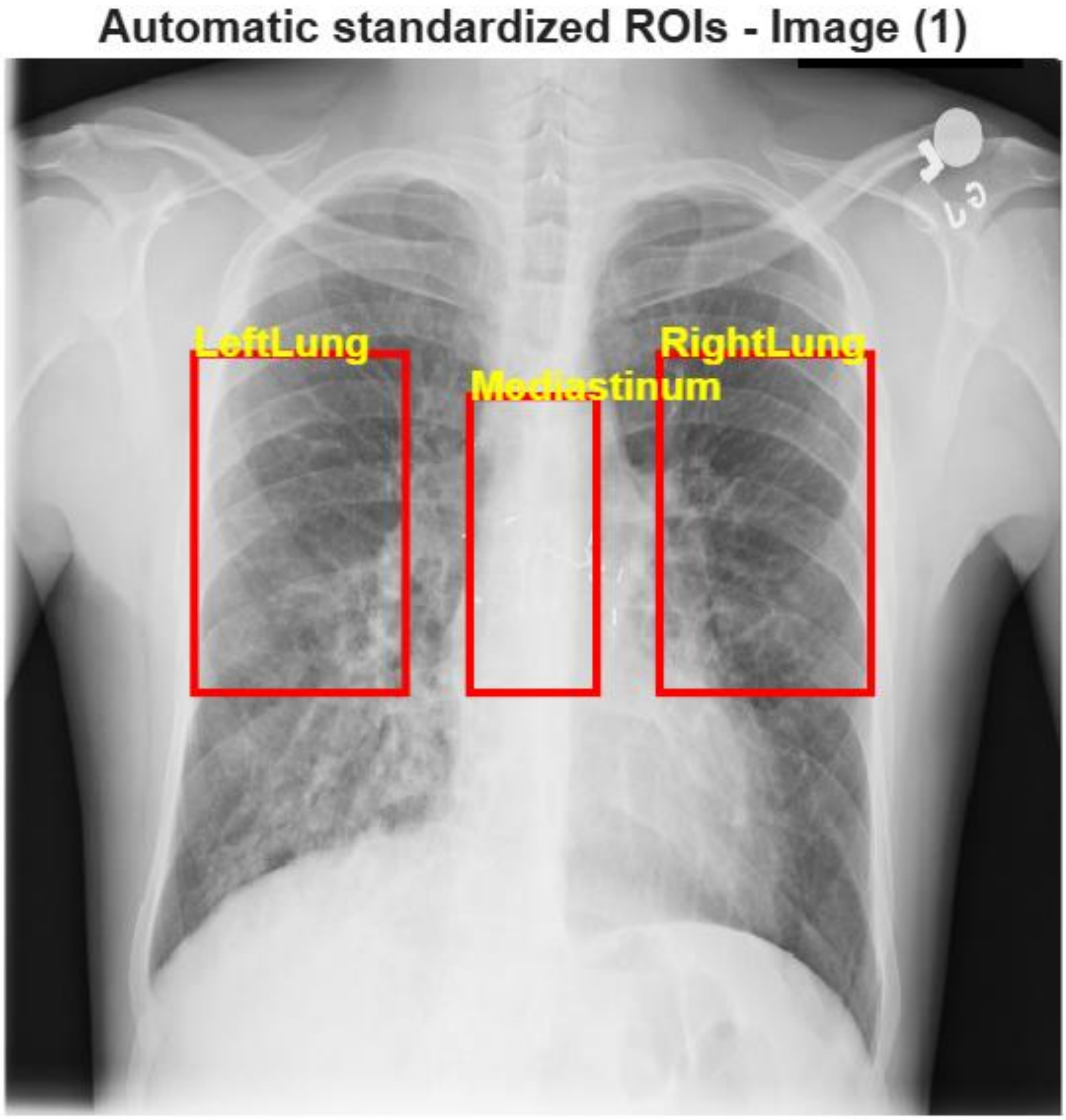

Figure 1. Standardized automatic ROIs on representative Image (1).

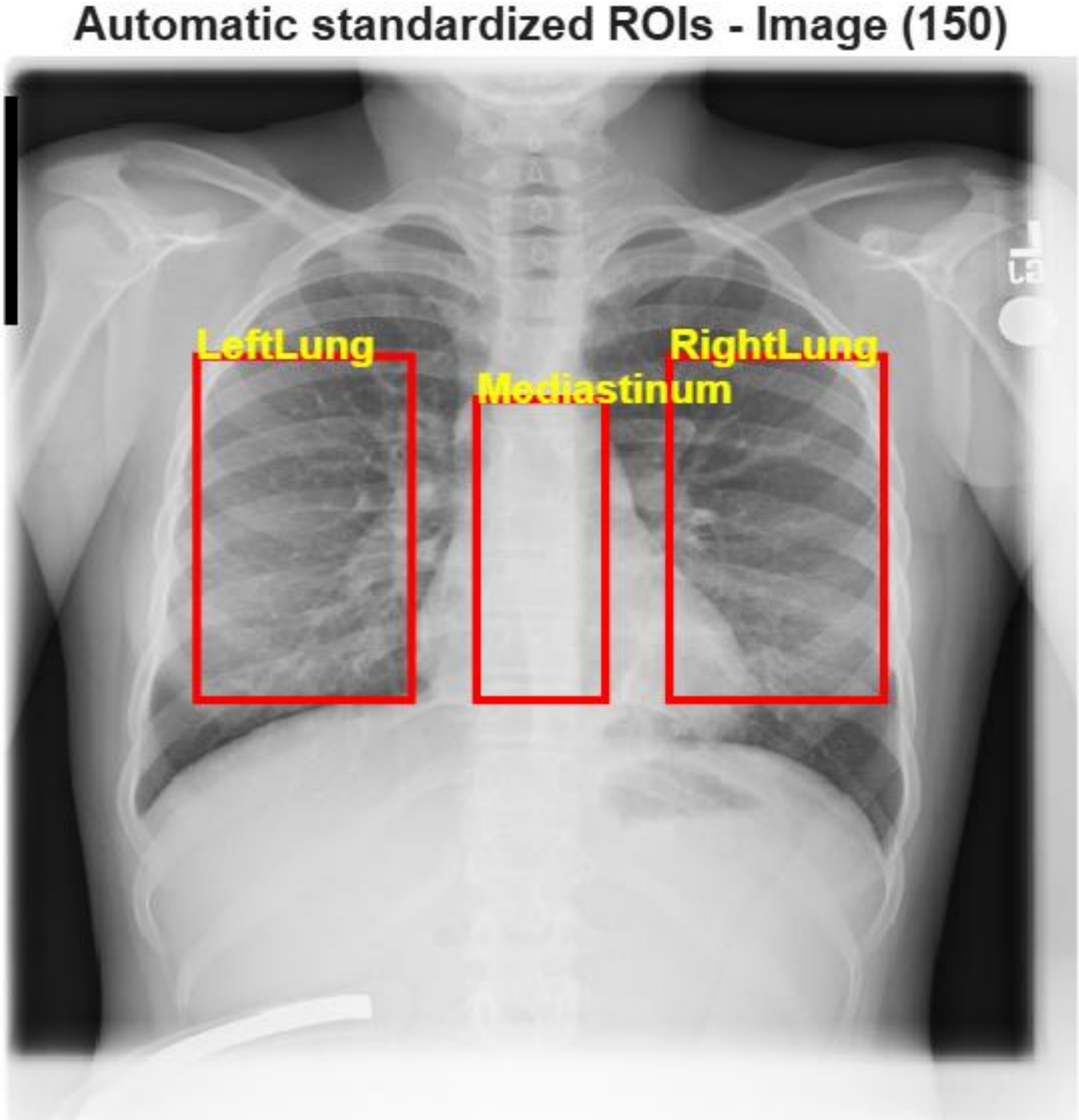

Figure 2. Standardized automatic ROIs on representative Image (150).

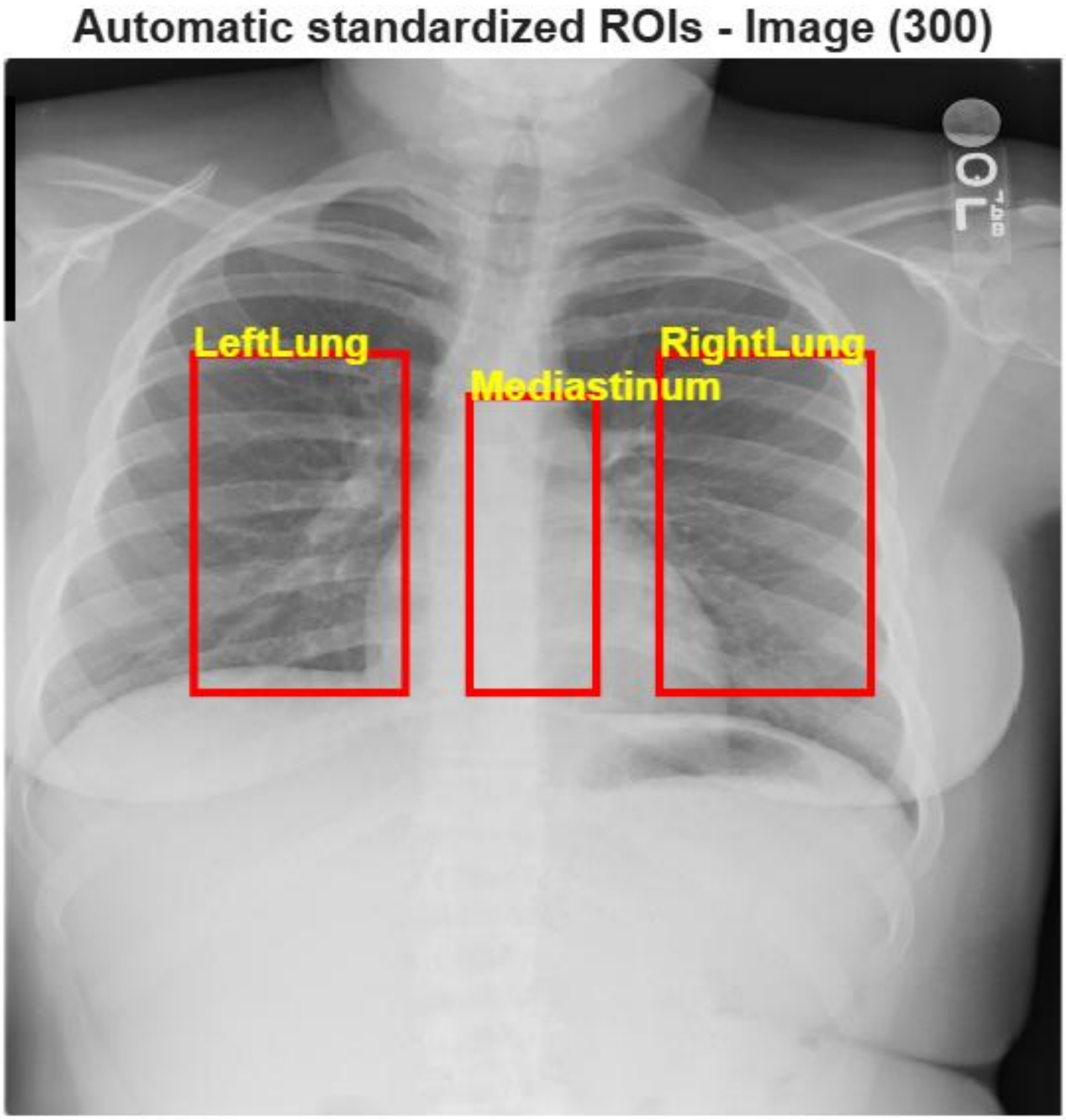

Figure 3. Standardized automatic ROIs on representative Image (300).

3. Quantitative Results

Table 1 summarizes the 300-image evaluation. Values are reported as mean ± standard deviation.

Dose	Condition	RMSE	PSNR (dB)	SSIM	Entropy	EME
10%	Baseline	0.02502 ± 0.00132	32.05 ± 0.49	0.6651 ± 0.0325	7.385 ± 0.189	11.511 ± 3.806
10%	Enhanced	0.06364 ± 0.04684	25.44 ± 4.79	0.8438 ± 0.0492	7.518 ± 0.059	13.945 ± 2.543
25%	Baseline	0.01583 ± 0.00084	36.03 ± 0.50	0.8244 ± 0.0193	7.345 ± 0.195	10.721 ± 3.796
25%	Enhanced	0.06283 ± 0.04710	25.62 ± 4.93	0.8904 ± 0.0455	7.501 ± 0.058	13.673 ± 2.476
50%	Baseline	0.01119 ± 0.00059	39.03 ± 0.50	0.9016 ± 0.0114	7.325 ± 0.198	10.364 ± 3.795
50%	Enhanced	0.06254 ± 0.04719	25.68 ± 4.98	0.9093 ± 0.0442	7.490 ± 0.058	13.558 ± 2.462

Table 2. Relative enhancement changes compared with the unenhanced low-dose baseline.

Dose	SSIM change	EME change	Entropy change	Pixel-fidelity trend
10%	+26.9%	+21.1%	+1.8%	RMSE ↑ / PSNR ↓
25%	+8.0%	+27.5%	+2.1%	RMSE ↑ / PSNR ↓
50%	+0.9%	+30.8%	+2.3%	RMSE ↑ / PSNR ↓

3.1 Full-reference interpretation

The structural effect is dose-dependent. At 10% dose, the enhancement chain increased mean SSIM from 0.6651 to 0.8438, an improvement of approximately 26.9%. At 25% dose, SSIM increased from 0.8244 to 0.8904 (about 8.0%). At 50% dose, the gain was modest, from 0.9016 to 0.9093 (about 0.9%). This pattern is sensible: when the input is already relatively high quality, there is less structural degradation to recover.

RMSE and PSNR moved in the opposite direction. For example, at 25% dose RMSE increased from 0.01583 to 0.06283 and PSNR decreased from 36.03 dB to 25.62 dB. This should not be presented as a contradiction. Histogram matching and enhancement deliberately remap gray levels, so the output may become structurally more similar while no longer matching the original pixel intensities as closely. The final chain therefore optimizes structural/perceptual restoration rather than strict least-squares fidelity.

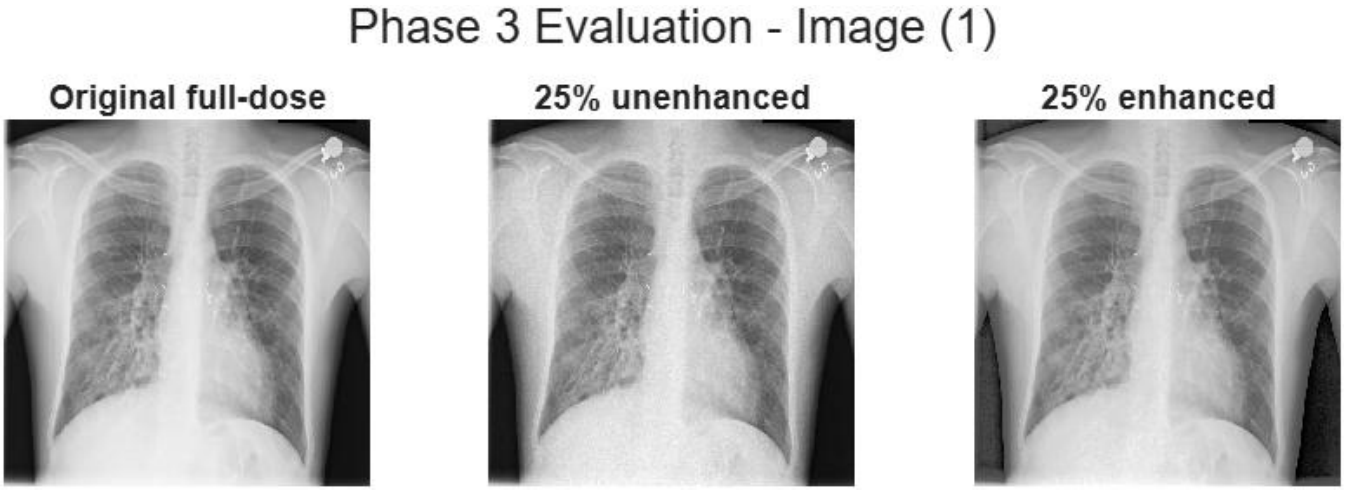

Figure 4. Phase 3 visual comparison for Image (1): original full-dose, 25% unenhanced, and 25% enhanced.

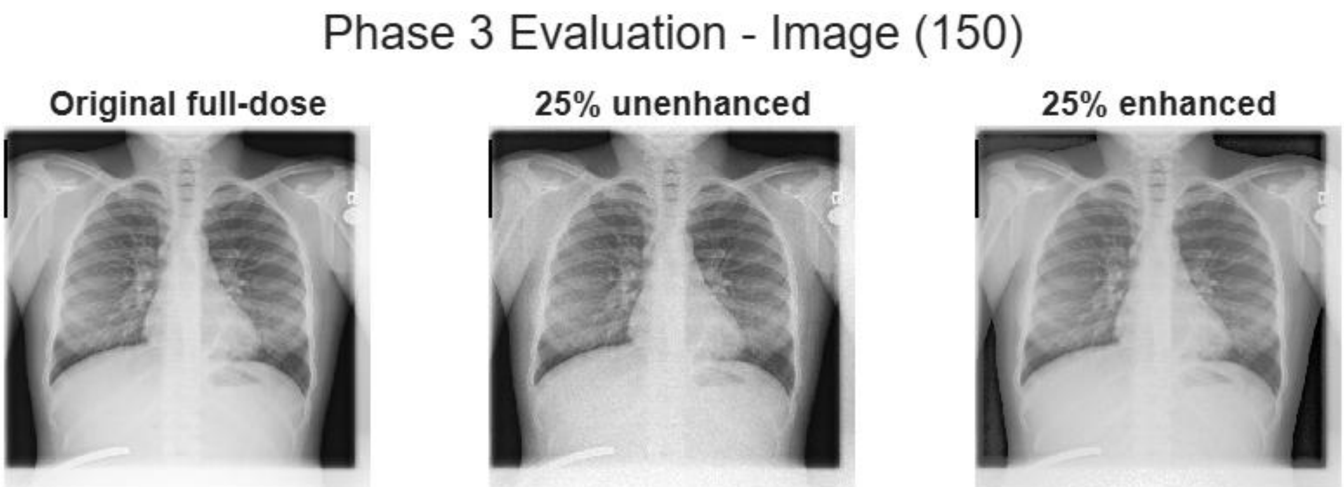

Figure 5. Phase 3 visual comparison for Image (150).

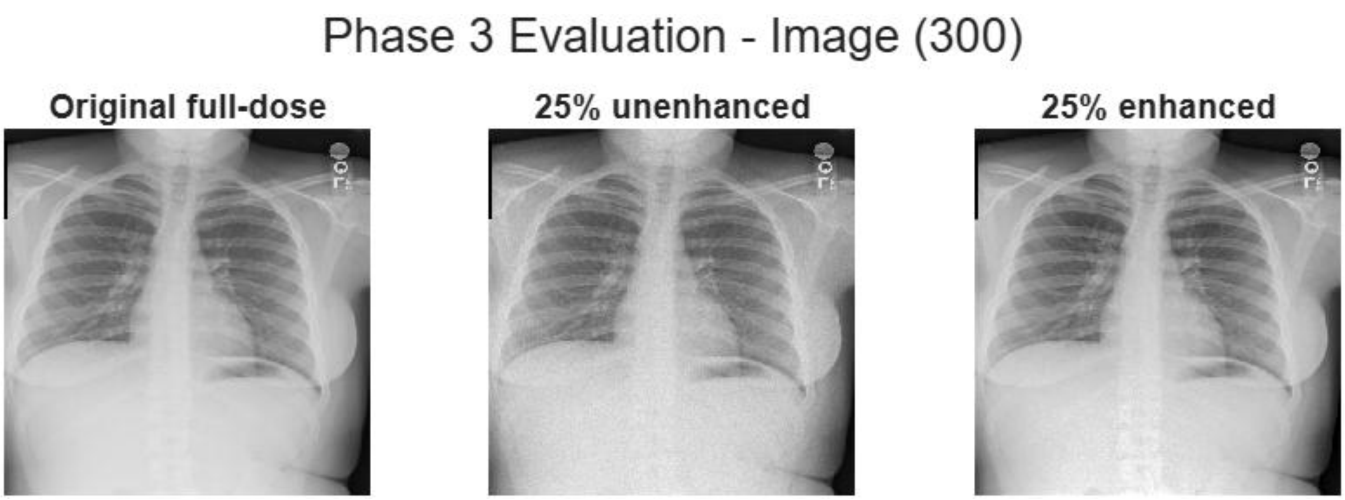

Figure 6. Phase 3 visual comparison for Image (300).

4. No-Reference Enhancement Analysis

Enhancement increased EME at every dose level: from 11.511 to 13.945 at 10%, from 10.721 to 13.673 at 25%, and from 10.364 to 13.558 at 50%. The relative increase was largest at 50% because the baseline EME was lowest, but this does not mean the 50% condition benefited most clinically. The full-reference SSIM results show that the major structural benefit occurred at 10% dose.

Entropy also increased slightly after enhancement at all dose levels. Because entropy can rise due to either useful detail or noise, it was interpreted jointly with SSIM and EME rather than used as a stand-alone quality score. The combination of increased SSIM and EME, particularly at 10% and 25% dose, supports the conclusion that the enhancement chain restored meaningful structural contrast rather than only increasing global intensity variability.

Interpretation rule Higher entropy or EME alone is not sufficient evidence of better medical-image quality. The most convincing result is improvement in structural similarity together with controlled visual appearance and independent human preference.

5. CLAHE Clip-Limit Sensitivity

A sensitivity sweep was performed over the custom CLAHE clip limit. The analysis compared normalized EME contrast gain with a normalized noise-amplification measure. The detected crossover occurred at a clip limit of 0.0025, which was the first tested operating point where the noise-amplification curve exceeded the contrast-gain curve.

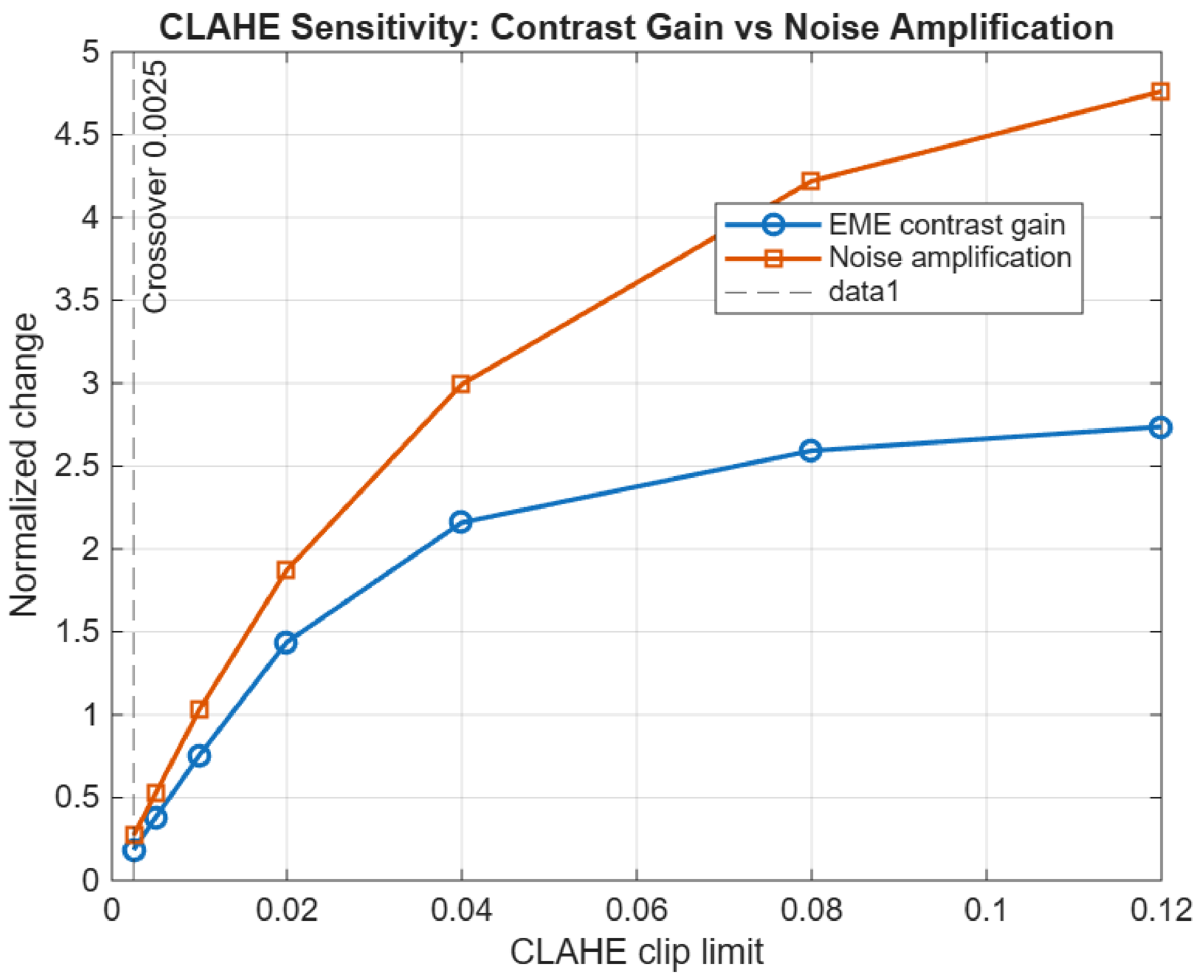

Figure 7. CLAHE sensitivity: normalized contrast gain versus noise amplification as clip limit increases.

The graph shows that both contrast gain and noise amplification increase as the clip limit becomes less restrictive, but noise grows more quickly than EME gain. Since the crossover already occurs at the smallest tested clip limit (0.0025), the tested CLAHE configuration is noise-dominated over the full evaluated range according to this criterion. This supports the Phase 2 decision to use histogram matching rather than CLAHE in the selected final chain.

Sensitivity conclusion For the tested image set, tile configuration, dose condition, and normalization method, increasing the CLAHE clip limit produced disproportionate noise amplification. A lower clip limit than 0.0025 would be needed to determine whether a truly contrast-dominant operating region exists.

6. Structured Visual Assessment

The project requires 20 randomized enhanced/unenhanced pairs to be evaluated by at least three colleagues. The MATLAB workflow generates blinded A/B pairs and a separate answer key. At the time of this report, numerical image processing was complete, but the human reviewer study had not yet been completed. Therefore, preference rates are intentionally not reported.

Visual-assessment item	Status / requirement
Number of randomized pairs	20
Minimum reviewers required	3
Current status	Pending reviewer responses
Planned response	A / B / Tie
Primary outcome	Enhanced-image preference rate (%)

When reviewer responses are available, the separate reviewer-analysis script can score each response against the hidden randomization key and report overall preference, reviewer-specific preference, and dose-specific preference without repeating the 300-image numerical analysis.

7. Implementation Compliance and Limitations

The project specification requires histogram equalization, histogram matching, CLAHE clipping/redistribution, adaptive median filtering, unsharp/high-boost filtering, and convolution kernels to be implemented from scratch. The Phase 2/3 workflow was structured accordingly, while MATLAB library equivalents were reserved for correctness checks. The current Phase 3 report focuses on the final-chain evaluation and CLAHE sensitivity outputs supplied from MATLAB.

The main methodological deviation is the ROI procedure. The project asks for manually placed ROIs, whereas the executed analysis used automatic standardized boxes. This improves reproducibility and eliminates operator dependence, but it can reduce anatomical specificity. Consequently, any ROI-based local-contrast conclusions should be interpreted as standardized regional measurements, not as expert anatomical annotations.

A second limitation is that the NIH ChestX-ray14 PNG files are post-processed 8-bit images. The Poisson model is therefore a controlled approximation of quantum-noise degradation rather than a scanner-calibrated physical dose model. Finally, the human visual-preference experiment remains incomplete and must be added before the project fully satisfies the Phase 3 brief.

8. Conclusion

Phase 3 demonstrates that the selected enhancement chain changes image quality in a clinically relevant but metric-dependent way. Enhancement consistently improved structural similarity, especially at severe dose reduction, and increased local contrast as measured by EME. The strongest structural recovery occurred at 10% dose, where SSIM rose from 0.6651 to 0.8438. At the same time, RMSE increased and PSNR decreased because the enhancement process intentionally remapped image intensities. The output should therefore be described as structurally and perceptually enhanced rather than as a pixel-perfect reconstruction of the original full-dose image.

The CLAHE sensitivity experiment further showed that noise amplification overtook EME contrast gain at the lowest tested clip limit of 0.0025 and remained dominant as the clip limit increased. This result argues against aggressive CLAHE for these simulated Poisson-degraded radiographs and supports the use of a more controlled contrast-restoration stage. Final completion of Phase 3 requires only the blinded human preference study with at least three reviewers and subsequent reporting of preference rates.

Overall Phase 3 outcome Numerical evaluation is complete for 300 images. The enhancement chain improves SSIM and EME, with the largest structural benefit at 10% dose. The remaining required task is the 20-pair, three-reviewer visual assessment.

Overall Project Conclusion

Across the three phases, the project established a consistent acquisition-to-restoration-to-evaluation workflow for low-dose chest radiography. Phase 1 quantified how spatial sampling, gray-level precision, and photon statistics degrade image quality. Phase 2 selected a controlled enhancement chain consisting of Gaussian denoising, histogram matching, and conservative high-boost filtering. Phase 3 showed that this chain improves structural similarity and local enhancement most strongly under severe dose reduction, while also demonstrating the expected trade-off between perceptual/structural restoration and strict pixel-wise fidelity. The results support careful, controlled enhancement rather than aggressive local contrast amplification. Completion of the blinded human preference study remains the final requirement for a fully closed Phase 3 evaluation.

Appendix A — Numerical Results Used in This Report

Dose	Condition	RMSE	PSNR (dB)	SSIM	Entropy	EME
10%	Baseline	0.02502 ± 0.00132	32.05 ± 0.49	0.6651 ± 0.0325	7.385 ± 0.189	11.511 ± 3.806
10%	Enhanced	0.06364 ± 0.04684	25.44 ± 4.79	0.8438 ± 0.0492	7.518 ± 0.059	13.945 ± 2.543
25%	Baseline	0.01583 ± 0.00084	36.03 ± 0.50	0.8244 ± 0.0193	7.345 ± 0.195	10.721 ± 3.796
25%	Enhanced	0.06283 ± 0.04710	25.62 ± 4.93	0.8904 ± 0.0455	7.501 ± 0.058	13.673 ± 2.476
50%	Baseline	0.01119 ± 0.00059	39.03 ± 0.50	0.9016 ± 0.0114	7.325 ± 0.198	10.364 ± 3.795
50%	Enhanced	0.06254 ± 0.04719	25.68 ± 4.98	0.9093 ± 0.0442	7.490 ± 0.058	13.558 ± 2.462

Source: MATLAB Phase 3 console output supplied by the project author. GroupCount = 300 for each dose × condition combination.

Appendix B — Items Pending for Final Submission

- Complete the 20 randomized A/B pair assessment with at least three independent reviewers.
- Run the separate reviewer-analysis script and insert overall, per-reviewer, and per-dose preference rates.
- If strict compliance with the assignment is required, repeat local-contrast measurements using manually placed ROIs or explicitly obtain instructor approval for standardized automatic ROIs.
- Insert the library-equivalent correctness-check appendix table if required by the marking rubric.

9. References

1. NIH Clinical Center. NIH ChestX-ray14 dataset. Primary distribution: https://nihcc.app.box.com/v/ChestXray-NIHCC

2. Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., and Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

3. MATLAB Image Processing Toolbox documentation: image resizing, image-quality metrics, and image visualization functions.